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B.Sc. in SWE 6th Semester | 12 April 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION | **SUMMER SEMESTER, 2020-2021**
DURATION: 3 HOURS | **FULL MARKS: 150**

CSE 4617: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **any 6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. a) What are the following environment properties of a chess game with a clock: observability, determinism, episodic, static, and discrete? Given the following initial and goal states in Figure 1 of an 8-puzzle problem. Perform an A* search starting from the initial state to find the goal state. Stop the search as soon as the search expands a state at level three (the start state belongs to level 0). Use the "the number of misplaced tiles" as a heuristic to guide the search. 4+8 (CO1, CO3) (PO1, PO2)

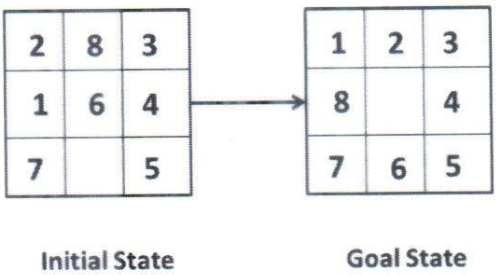
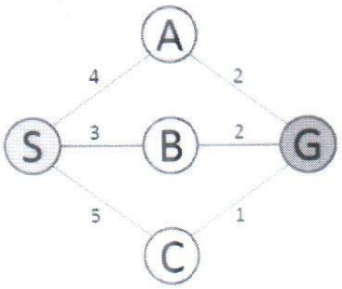


Figure 1: Initial and goal states of an 8-puzzle problem.

b) "We need to trade-off the accuracy of heuristic against the overhead for computing that heuristic on every search state" – Justify the statement. Step-wise perform SMA* search (memory: 3 nodes) on the following graph in Figure 2, where **S** and **G** are the start and goal nodes respectively. The heuristic value of each node is given in the table underneath. 5+8 (CO2) (PO2)



	S	A	B	C	G
heuristic	3	0	2	1	0

Figure 2: Graph and table for question 1.b)

2. a) Describe the terms 'plateau' and 'ridge' in respect of a state space landscape. Briefly explain different types of mutation operations in genetic algorithms. 4+6 (CO1) (PO1)

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b) "Starting from a randomly generated 8-queens state, steepest-ascent hill climbing gets stuck 86% of the time, solving only 14% of problem instances" – but is there any good side of it, and how can we improve this scenario? Perform a local search up to 5 steps on an 8-queens problem starting from an initial configuration as shown in the Figure 3 below (the numbers represent the total conflicts against respective queens). Here, each queen is dedicated to its existing column, and one single (usually the most aggressive) queen can be moved at a time. 7+8 (CO2, CO3) (PO2, PO4)

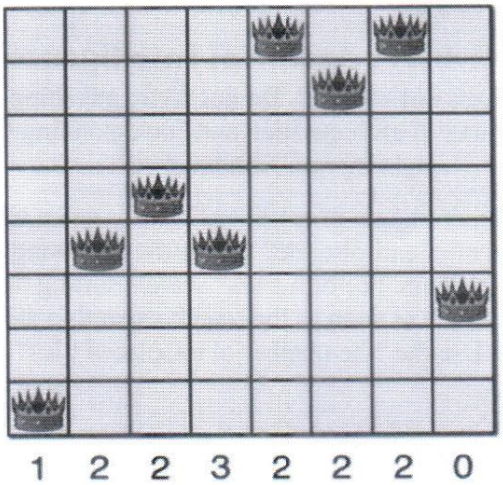

Figure 3: 8-queens problem for question 2.b)

3. a) What is higher order constraint in a constraint satisfaction problem (CSP)? What is Thrashing problem in Backtracking search? How can you avoid that? Explain with an example. 2+4+4 (CO1) (PO1)

b) Encode the following graph colouring problem, given in Figure 4, into a constraint satisfaction problem (CSP) having a finite set of variables, a finite domain for each variable, and a set of constraints. In this problem each vertex of the following graph has to be coloured by either red, green, or blue, such that no two adjacent vertices share the same colour. Solve the problem using backtracking considering the order in which you choose variables and their values, which may have a huge impact on search space size. Apply arc-consistency for earlier detection of failure. 15 (CO3) (PO3)

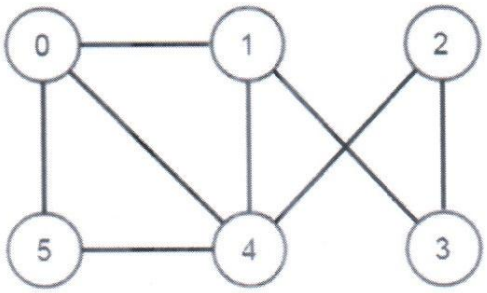

Figure 4: Graph for question 3.b)

4. a) How can you calculate the heuristic value of a state in the tic-tac-toe game? Explain with necessary diagram how you can apply the two-ply minimax algorithm to the opening move of the tic-tac-toe. 4+8 (CO3) (PO3)

b) How can you exploit the following two facts of an adversary: 4 (CO1) (PO2)

i) If a position is provably bad

ii) If the adversary can force a bad position?

c) Given the following search tree in Figure 5, step-wise show how the alpha and beta cuts can be applied to the tree using alpha-beta pruning technique, and which parts of the search tree are pruned as a result. 9 (CO2) (PO4)

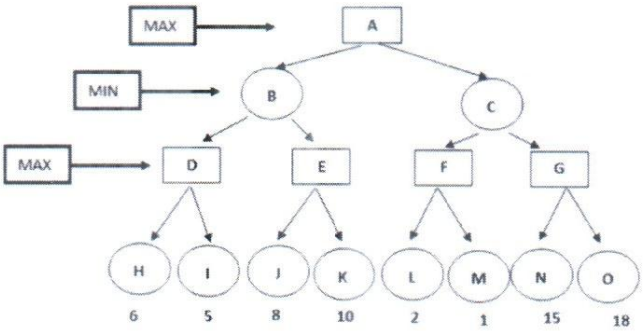

Figure 5: Search tree for question 4.c)

5. a) What is meant by conjunctive normal form (CNF) in propositional logic? Find out whether the following statements are valid, satisfiable, or unsatisfiable. 3+9 (CO1) (PO1)
   i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$
   ii. $(A \vee B) \wedge \neg(A \Rightarrow B)$
   iii. $(Smoke \Rightarrow Fire) \Rightarrow (\neg Smoke \Rightarrow \neg Fire)$

   b) What are the criteria for a knowledge base to be in horn form? Given the following Knowledge Base (KB). Prove that "It will not rain" by converting the KB into its Horn form in propositional logic, and then by applying forward chaining. 3+10 (CO1, CO2) (PO1, PO2)
   i. *The humidity is high or the sky is cloudy.*
   ii. *If the sky is cloudy, then it will rain.*
   iii. *If the humidity is high, then it is hot.*
   iv. *It is hot.*

6. a) Explain the advantages of first-order logic (FOL) over propositional logic (PL). Illustrate the common mistakes to avoid in universal and existential quantification. 5+5 (CO1) (PO1)

   b) What is generalized modus ponens? Given the following sentences: 2+13 (CO1, CO3) (PO1, PO3)
   i. *Jack owns a dog.*
   ii. *Every dog owner is an animal lover.*
   iii. *No animal lover kills an animal.*
   iv. *Either Jack or Jill killed the cat, who is named Tuna.*

   Now, write them in first-order logic (FOL), convert the FOL sentences into an inferentially equivalent sentence in conjunctive normal form (CNF), and finally prove, by resolution refutation (contradiction), that *Jill kills the cat.*

7. a) What is the purpose of causal links in partial plan representation? Step-wise generate a partially ordered plan of the "Wearing Socks and Shoe" problem, where the problem allows four actions, namely *left-sock* (pre: CleanLeftSock; add: LeftSockOn), *right-sock* (pre: CleanRightSock; add: RightSockOn), *left-shoe* (pre: LeftSockOn; add: LeftShoeOn), *right-shoe* (pre: RightSockOn; add: RightShoeOn). The initial state represents {CleanLeftSock, CleanRightSock} and the goal requires LeftShoeOn and RightShoeOn. Explain how a causal link is protected during the above partial plan generation if the *right-sock* action additionally removes the atom LeftSockOn. 2+10 (CO1, CO3) (PO1, PO3)

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b) Given a four knights problem in a 3×3 grid board, where the goal is to get each knight into the opposite corner. (A knight can move to an unoccupied space according to an L-shape movement: it must move two spaces horizontally and one vertically, or two spaces vertically and one horizontally.) 8+5 (CO4) (PO3, PO4)

i. Represent the problem in classical planning representation by writing a domain file and a problem file in PDDL.

ii. What could be an admissible heuristic for this problem? Write a solution (as sequence of actions) according to your domain action descriptions, and prove the correctness of the solution by showing that successively applying that action sequence leads the initial state to a goal state.

Initial positions of the four knights 1..4:

1 _ 2

_ _ _

3 _ 4

Goal positions of those four knights:

4 _ 3

_ _ _

2 _ 1

8. a) What is the PDDL expression of the statement, "A people *p* is not healthy or the people *p* is playing" using the "imply" keyword? Explain Sussman anomaly with respect to a block world problem. 4+7 (CO1) (PO1)

b) Distinguish between admissible and consistent heuristics. Given a set of seven coins, a player takes a set and divides it into two unequal sets. The player who cannot do uneven split, looses. Give an explicit description of the problem showing the complete state space, actions, and goals. 6+8 (CO1, CO2) (PO1, PO2)